

Scalability & Future Plan

Date	30 June 2026
Team ID	XXXXXX
Project Name	Rising Waters – AI-Powered Flood Prediction and Early Warning System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a static weather dataset only	Prediction accuracy can be improved with real-time APIs integration	High
2	Excel-based local storage	Scale is limited to local reads and writes	Medium
3	Basic email/alert system	Emergency alerts are not automated to SMS gateways	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local development users	Already deployed on Vercel cloud architecture
2	Data Storage	Excel dataset sheet	Integrate relational databases like PostgreSQL or MySQL
3	Performance	XGBoost Classifier model	XGBoost is implemented with 96.55% validation accuracy
4	Security	Basic Flask input validation	Add authentication and role-based access for disaster response teams

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Improve UI/UX	1 Month	Better user experience
Phase 2	Add Country Search Feature	2 Months	Easier prediction process
Phase 3	Deploy on Cloud	3 Months	Public accessibility
Phase 4	Dashboard & Analytics	4 Months	Interactive visual insights